

From Human-AI Performance to Sustained Capability

Toward a Design Science of Human-AI Developmental Ecologies

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Abstract

Artificial intelligence is arriving amid a period of unusually dense and interacting global challenges: ecological disruption, geopolitical instability, institutional fragility, economic transformation, information disorder, technological acceleration, and persistent failures of collective coordination. These problems differ in their causes and mechanisms, but together they pose a recurring civilizational question:

How can human beings and human systems become more capable of responding wisely to the conditions they are creating?

The language of polycrisis is itself contested, but a growing systems literature examines how crises can become causally entangled through common stresses, cascading effects, and cross-system feedbacks (Lawrence et al., 2024). Resilience and adaptive-governance traditions likewise emphasize the capacity of social-ecological systems to reorganize under disturbance and the importance of interactions among people, organizations, institutions, and levels of governance (Walker et al., 2004; Folke et al., 2005).

Many traditions have addressed parts of this problem through consciousness, human development, sustainability, education, institutional reform, deliberation, intentional community, and technological design. Their limitations are not uniform, but several recurring difficulties appear. Some offer profound meaning but weak causal traction. Some generate concrete local benefit without mechanisms capable of propagating broadly. Some address development without materially changing the systems within which human capacities are exercised. Others engage civilizational problems while offering relatively weak leverage over the infrastructures increasingly organizing social life.

Artificial intelligence is distinctive because it cuts across these levels simultaneously. AI increasingly participates in cognition, learning, scientific inquiry, economic production, institutional decision-making, education, culture, communication, governance, and collective problem-solving. Economic and organizational research already treats AI as consequential not only for productivity but for institutional structure, decision-making power, information flows, human wellbeing, and transition dynamics (Brynjolfsson, Korinek, & Agrawal, 2025; Bankins et al., 2026).

It is not merely another proposal for how people should think or behave. It is becoming part of the environments through which people think, learn, decide, relate, organize, create knowledge, and exercise agency. AI can therefore be understood not only as a new technological capability but as a historically consequential intervention into human and collective capability.

At the same time, human-AI research is moving beyond model-centered evaluation. Recent reviews identify interaction design, changing human-machine relationships, organizational and societal implications, multilevel effects, and the need for more longitudinal and context-sensitive research as important frontiers, while continuing to describe theoretical fragmentation and limitations in integrative evaluation (Seini, Adam, & Preko, 2026; Bankins et al., 2026). Post-deployment monitoring research similarly identifies human-AI feedback loops, beneficial-impact measurement, and validated real-world monitoring methods as unresolved methodological problems (Rao et al., 2026).

The emerging problem is therefore not that these dimensions have been overlooked. It is that recognition of them does not yet provide a cumulative account of how nested developmental ecologies organize local human-AI configurations, how specified participation protocols become enacted participation topologies, how consequential operations are recurrently distributed, and how those patterns alter capability across levels and time.

This paper develops a prospective research architecture organized around that problem. It distinguishes present performance from sustained capability; treats the distribution of consequential operations as a central substructure of enacted participation; separates ecological conditions, local configuration, intended protocol, and observed topology; and asks how recurrent participation changes future realizable participation across microgenetic, mesogenetic, and longitudinal timescales.

The Human-AI Developmental Research program (HAIDR) is presented not as a comprehensive theory of human-AI effects, but as an integrative architecture for connecting nested ecology, local configuration, participation protocol, enacted topology, candidate developmental mechanisms, capacity location, human-AI pathways, and observed trajectories across human, coupled, collective, and institutional levels.

The longer-term ambition is a design science capable of asking:

Which human-AI configurations and participation protocols, enacted as which participation topologies, produce which capability trajectories, for whom, under what conditions, across which levels and timescales?

1. From Artificial Capability to Sociotechnical Trajectory

Contemporary AI evaluation already extends well beyond model benchmarks. Human-AI research investigates teaming, function allocation, cognitive offloading, learning and skill, organizational adoption, agentic oversight, collective effects, and post-deployment behavior. Recent interdisciplinary reviews explicitly describe the field as multilevel and context-dependent while also finding substantial fragmentation across disciplinary and analytical levels (Seini et al., 2026; Bankins et al., 2026).

The scientific transition is therefore not adequately characterized as a move from a field concerned only with model capability toward one newly discovering human and social effects.

A more accurate transition is: **from increasingly sophisticated but partially separated forms of evaluation toward cumulative inference across configurations, levels, and timescales.**

Several empirical and theoretical literatures already establish pieces of this problem.

Human-AI performance is highly heterogeneous: a preregistered meta-analysis of 106 experiments found that human-AI combinations improved on humans alone on average, but did not on average

outperform the better of humans or AI alone, with outcomes varying substantially by task and relative capability (Vaccaro, Almaatouq, & Malone, 2024).

Classic automation research established much earlier that automation does not merely substitute for human activity; it reorganizes what humans do and creates new coordination and supervisory demands (Parasuraman, Sheridan, & Wickens, 2000).

Organizational reviews now similarly emphasize that the effects of AI depend on interactions among individual, group, organizational, and broader contextual factors rather than on technological capability alone (Bankins et al., 2024, 2026).

And post-deployment evaluation remains comparatively immature: NIST's 2026 review describes validated monitoring methodologies and common terminology as nascent and scattered, while specifically identifying insufficient research on human-AI feedback loops and metrics for beneficial impacts to humans (Rao et al., 2026).

We therefore increasingly know that configuration matters. What remains harder is connecting these observations strongly enough to answer:

Which configurations cause which durable consequences, under what conditions, through which mechanisms, and with what degree of transportability?

This difficulty is especially important because conclusions at one level do not automatically license conclusions at another.

- Improved task performance does not establish increased human competence.
- Higher individual productivity does not establish greater collective intelligence.
- Organizational throughput does not establish preserved expertise or resilience.
- Formal human accountability does not establish practical capacity to intervene.
- And successful AI applications to individual social problems do not establish increased societal adaptive capacity.

The distinction between levels is not merely theoretical. For example, recent meta-analytic evidence suggests that generative AI can improve local creative production while simultaneously producing a small but significant homogenization of outputs across participants, illustrating how individual and collective outcomes can diverge (de Rooij & Biskjaer, 2026).

The resulting frontier is therefore a problem of sociotechnical evaluation across levels and time.

The relevant system increasingly includes not simply the model but the human participants, local configuration, intended participation protocol, enacted workflow, organization, incentives, institutions, infrastructures, and histories through which model capabilities are realized.

The central research question becomes:

How do configurations of artificial capability, human participation, interaction design, delegation, institutions, and history alter human and collective capability over time?

For HAIDR, the longitudinal system under study is an evolving coupled human-AI developmental system embedded within an evolving developmental ecology. A human-AI pathway is the ecology-conditioned longitudinal unfolding of developmental possibilities; a developmental trajectory is the empirically observed time-extended course of change within that pathway. The distinction separates what is evolving, what the program seeks to explain, and what is observed over time.

2. Human–AI Developmental Ecologies

I use **human-AI developmental ecology** to refer to *the nested, historically conditioned field of material, political-economic, societal-cultural, institutional, organizational, task/social, and proximal human-AI conditions through which recurrent participation occurs and through which developmental consequences become possible*. The ecology is not merely background context: it is part of the causal organization of participation.

This is a synthetic construct rather than a claim that the field lacks adjacent concepts. Distributed-cognition research has long argued that cognition should sometimes be analyzed across whole environments of people, artifacts, and technologies rather than confined to the individual (Hollan, Hutchins, & Kirsh, 2000). Cognitive-offloading research likewise examines how people alter the information-processing demands of tasks through external action and technological support (Risko & Gilbert, 2016). Human-factors research analyzes distributions of function between humans and automation, while Coactive Design emphasizes interdependence rather than treating autonomy as an isolated system property (Parasuraman et al., 2000; Johnson et al., 2014).

HAIDR's developmental-ecology proposal builds on these traditions while adding a longitudinal question:

How does repeated participation in a given sociotechnical ecology alter what its participants subsequently become capable of doing, and how do those changes alter the conditions of later participation?

The ecology is recursive. Participation can change the person, workflow, organization, or institutional arrangement; those changes alter subsequent participation conditions; and the resulting cycles can amplify, stabilize, displace, or reorganize capacities over time.

Local human–AI configuration

Within the wider ecology, the local human-AI configuration is the proximal, comparatively manipulable arrangement through which participation is instantiated. It includes model capability and modality, memory and continuity, retrieval architecture, interface and tools, scaffolding, autonomy and persistence, and other local system properties.

This distinction matters because a local configuration is not the whole developmental ecology. Two people can use technically similar configurations inside very different task, organizational, institutional, or political-economic conditions, and those wider conditions can select, reward, constrain, or stabilize different forms of participation.

Interaction / participation protocol

The interaction or participation protocol is the specified, designed, or prescribed organization of participation within a local configuration. It may structure initiative, turn and timing rules, scaffolding, feedback, epistemic authority, task decomposition, provenance requirements, verification, revision, and scaffold withdrawal.

A developmental participation protocol is a subtype intentionally designed around developmental objectives. The protocol describes intended organization; it does not establish what actually occurred.

Enacted participation topology

Participation topology is the actual relational organization of consequential activity enacted during participation. It records who initiates, retrieves, generates, interprets, plans, judges, monitors, detects error, corrects, decides, acts, and determines completion; when those operations occur; who has authority to redirect them; how information is sourced; and how revision and recurrence are organized.

Authority matters as much as activity. A human may nominally review a decision while possessing little effective opportunity or competence to alter it. Recent research on software-agent oversight similarly finds that practical oversight comprises several different forms of work - including a priori control, co-planning, real-time monitoring, and post hoc review - rather than a single generic "human-in-the-loop" function (Dhanorkar, Passi, & Vorvoreanu, 2026). Formal human authority therefore does not establish effective human control.

Distribution of consequential operations

A central substructure of participation topology is the distribution of consequential operations: who actually does what. The ancestry here is explicit. Parasuraman et al. distinguish automation of information acquisition, information analysis, decision/action selection, and action implementation; Coactive Design emphasizes observability, predictability, directability, and interdependence in joint activity (Parasuraman et al., 2000; Johnson et al., 2014). HAIDR extends this logic longitudinally rather than replacing it.

The word *consequential* is deliberately neutral about value. An operation may be consequential because it shapes learning, judgment, agency, responsibility, or later strategy space; this does not predetermine whether its redistribution is beneficial or harmful.

Variable placement

The ontology creates a practical separation among kinds of variables. Model capability, modality, memory, retrieval architecture, interface, tools, scaffolding, and persistence belong primarily to local configuration. Rules for timing, initiative, feedback, authority, verification, retrieval, and withdrawal belong to protocol. Actual retrieval, generation, judgment, monitoring, correction, and action belong to enacted topology. Human readiness, expertise, goals, motivation, identity, metacognitive ability, and prior history remain participant variables that shape how a configuration and protocol are enacted.

Retrieval illustrates why the distinctions matter. Stored information may be available through a retrieval architecture; a protocol may specify when or how retrieval should occur; actual retrieval may or may not happen; retrieved information may or may not be used consequentially. Availability is not retrieval, retrieval is not consequential use, and consequential use is not itself a developmental effect.

Nested ecological levels

The local configuration is embedded within broader ecological levels that are analytically useful without being treated as exhaustive natural kinds:

- **material and ecological conditions** — energy, infrastructure, environmental constraints, resource flows, and externalities;
- **political-economic ecology** — ownership, market structure, labor conditions, distribution of benefits and costs, geopolitical power, and dependency;

- **societal-cultural ecology** — norms, media systems, knowledge institutions, inequality, public narratives, and patterns of participation;
- **institutional ecology** — professions, educational systems, law, regulation, standards, procurement, credentialing, liability, and accountability;
- **organizational ecology** — workflows, incentives, roles, metrics, time pressures, local authority structures, and coordination routines;
- **task and social ecology** — the immediate problems, uncertainty, stakes, collaborators, peers, mentors, and social expectations surrounding participation;
- **local human-AI configuration** — the proximal technical and procedural arrangement through which participation is instantiated.

Time and participation history cut across these levels rather than constituting another nested level. Prior adaptations, model changes, accumulated dependence, scaffold fading, organizational learning, and institutional path dependence can all alter what the same nominal configuration means later.

The same specified protocol could therefore plausibly produce different enacted topologies in a classroom designed around learning, a workplace optimized for throughput, an emergency-response environment, or a regulated clinical setting. This is not yet a settled empirical generalization; it is a design hypothesis implied by literatures showing that AI outcomes are contingent on organizational, task, user, and institutional conditions.

The governing safeguard is simple: **ecology and configuration condition participation; protocol structures intended participation; topology records enacted participation.** None should be inferred from another without evidence.

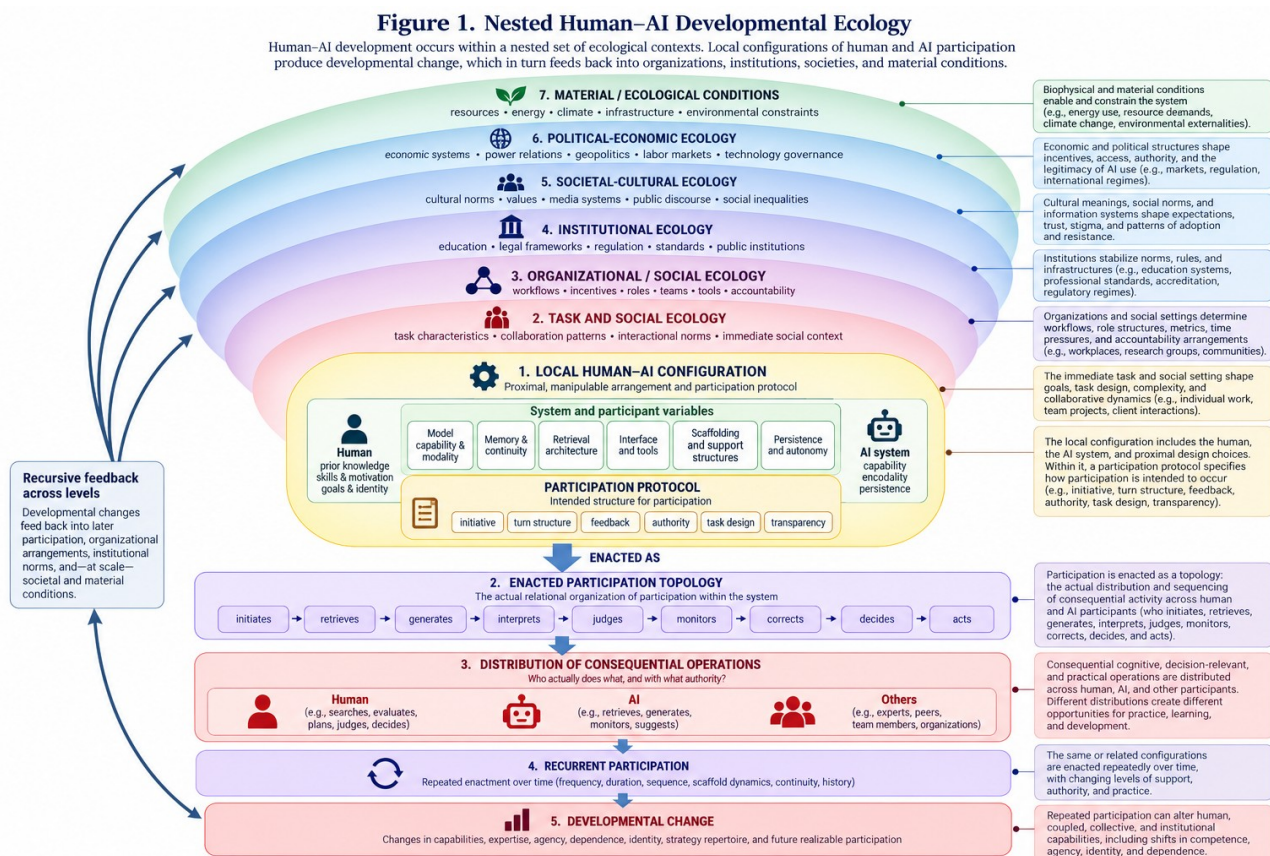


Figure 1. Human-AI development occurs within a nested developmental ecology. The local configuration contains the proximal technical arrangement and specified participation protocol; actual participation must be observed as an enacted topology, whose central substructure is the distribution of consequential operations. Developmental changes feed back into later participation and into organizational and institutional conditions.

3. Sustained Capability

The developmental framing introduces a distinction between **present performance** and **sustained capability**.

This distinction has strong intellectual ancestry. Learning science has long distinguished observable performance during acquisition from relatively durable learning expressed through retention and transfer. Improvements in immediate performance can occur without corresponding long-term learning, and the reverse can also occur (Soderstrom & Bjork, 2015).

Automation research similarly documents cases in which high automated performance can coexist with reduced human situation awareness or poorer ability to resume control after automation failure (Bainbridge, 1983; Endsley & Kiris, 1995).

Cognitive-offloading research complicates any simplistic conclusion that externalization is inherently harmful: offloading can reduce internal cognitive demand and can be adaptive, while its consequences depend on metacognition, task requirements, and the nature of the external support (Risko & Gilbert, 2016).

HAIDR therefore does not equate desirable development with maximization of unaided performance.

Two human-AI arrangements can produce comparable immediate results while generating different trajectories.

- One may strengthen independent judgment.
- Another may support stable complementary specialization.
- Another may remove opportunities through which expertise would otherwise develop.
- Another may weaken lower-level fluency while strengthening higher-order supervision.
- Another may increase organizational productivity while eroding the distributed expertise required for future correction.

“Augmentation” and “replacement” are therefore often too coarse. A more useful evaluation requires a multidimensional outcome vector distinguishing:

- present task performance;
- supported human-AI capability;
- retained human competence;
- practical intervention capability;
- higher-order orchestration;
- collective knowledge distribution;
- organizational capability;
- resilience under perturbation;
- infrastructure dependence;
- and changes in future strategy space.

The core evidential question becomes:

What observations, comparisons, causal assumptions, and validation standards justify the conclusion that a human-AI configuration produces sustained capability rather than merely strong present performance?

Existing literatures establish parts of this problem. The proposed contribution is their explicit conjunction under persistent AI participation. Unaided performance is therefore not a universal gold standard. Stable external support may constitute valuable capability. Some operations may reasonably migrate to artificial systems. Lower-level delegation may permit humans to develop new higher-order capacities.

The question is instead:

What capability changed, where does it now reside, what does it depend upon, and what happens when conditions change?

Coupled efficacy is therefore not equivalent to human development. A human-AI system can become more capable while a human capacity strengthens, weakens, moves to a higher-order supervisory role, or becomes dependent on continuing support. Capacity location must remain explicit.

This leads to a central integrative problem:

How does persistent redistribution of cognitive, agentic, and relational activity between humans and AI alter the future capacities and agency of the human, the capabilities of the coupled system, and the institutional conditions under which those trajectories become stabilized?

And then:

How should those divergent outcomes be evaluated and governed?

4. Participation History as a Developmental Variable

Nominal task allocation often conceals developmentally consequential differences. "Human review," for example, can mean independent judgment before AI exposure, judgment after AI recommendation, comparison between independently generated human and AI answers, targeted anomaly detection, substantive correction, procedural approval, or near-total deference accompanied by nominal sign-off.

Automation and human-factors research provides direct precedent for treating such differences as consequential. Bainbridge's "ironies of automation" highlighted the problem of leaving humans responsible for rare abnormal conditions while routine activity - and thus routine practice - is automated. Endsley and Kiris subsequently demonstrated poorer situation awareness and takeover performance under higher automation in an experimental navigation task (Bainbridge, 1983; Endsley & Kiris, 1995).

A developmentally informative participation history should therefore represent more than exposure volume or transcript length. At minimum it should distinguish four layers:

1. **ecological / configuration state** — which model, memory, retrieval, interface, tools, autonomy, scaffolding, task, and surrounding organizational conditions were present;
2. **intended protocol** — what rules structured initiative, timing, feedback, authority, verification, revision, and withdrawal;
3. **enacted participation** — what topology actually occurred, including the distribution of consequential operations;

4. **consequences and subsequent state** — what changed in performance, strategy, capacity, dependence, role, or later participation conditions.

Within the enacted layer, a useful representation may record:

*operation × actor × initiative × authority × sequence × timing of assistance × information
provenance × feedback × opportunity for revision × recurrence × persistence / background
activity*

This representation is a HAIDR proposal, not an established taxonomy. It extends earlier function-allocation and human-automation approaches by making longitudinal participation history explicit.

Several distinctions are especially important:

- **Initiative** — Who begins the cognitive or agentic operation?
- **Sequence** — Does human judgment occur before or after exposure to AI output?
- **Authority** — Who can meaningfully revise, reject, override, or redirect?
- **Epistemic provenance** — Are multiple judgments genuinely independent or generated from a common artificial source?
- **Generative contribution** — Who generates hypotheses, distinctions, alternatives, interpretations, or plans?
- **Evaluative contribution** — Who compares alternatives, establishes criteria, identifies error, and determines sufficiency?
- **Corrective contribution** — Who notices failure and performs the work required to repair it?
- **Recurrence** — Which operations are repeatedly practiced, intermittently practiced, or effectively abandoned?
- **Scaffold dynamics** — Does support remain constant, increase, fade, or adapt to demonstrated competence?
- **Transfer opportunity** — Does the ecology require operations to be enacted under new conditions, with reduced support, or outside the original coupling?
- **Temporal initiative** — Who carries prospective control across steps and intervals, including maintaining goals, noticing that action is needed, resuming work, monitoring progress, and preserving completion conditions?

Persistent agents make the final distinction especially important. If an artificial participant can preserve or recover state and continue, initiate, monitor, or revise goal-directed activity between explicit user interactions, the interaction boundary no longer marks the developmental-activity boundary. Between-session events therefore belong in the participation history when they alter consequential operations, authority, or future conditions.

Retrieval history also requires provenance. A stored memory, a retrieval opportunity, an actual retrieval event, and consequential use of retrieved information are different observations. Measurement should preserve those distinctions rather than treating memory or access as a scalar dose.

Participation-history measures earn their place only if they improve discrimination, prediction, intervention, or transportability. Trace is not coded construct, coded construct is not developmental inference, and a high-quality AI-assisted downstream artifact is not by itself evidence of independently available human capacity.

This is the intended function of integrative work: not to construct another encompassing vocabulary, but to identify distinctions at the interfaces of established literatures that materially change empirical inference.

5. HAIDR as a Nested, Multiaxial Research Architecture

The Human-AI Developmental Research program can therefore be stated as:

HAIDR is a candidate integrative research architecture for investigating how developmental ecologies organize recurrent human-AI participation, how repeated participation forms, displaces, or reorganizes capacities, and how those changes alter human-AI pathways across human, coupled, collective, and institutional levels.

A simple linear representation is insufficient. HAIDR instead uses three interacting analytic dimensions:

1. ecological level;
2. developmental process;
3. timescale.

These dimensions are multiaxial rather than strictly orthogonal. They can interact, but they should not be collapsed: a higher ecological level is not simply a longer timescale, and an event at one developmental process stage does not automatically establish change at another.

Ecological level

At the outer levels are material/ecological, political-economic, societal-cultural, institutional, organizational, and task/social conditions. Within them sits the local human-AI configuration. This multilevel orientation has strong precedent in organizational AI scholarship and adaptive-governance research, although HAIDR's particular nesting and developmental interpretation are its own synthesis (Bankins et al., 2024, 2026; Folke et al., 2005).

Higher-order environments create selection pressure on which local configurations and protocols become feasible, profitable, legitimate, mandatory, sustainable, or normal. Local changes can in turn alter roles, workflows, organizational capabilities, norms, procurement practices, and institutional expectations. Aggregation alone, however, does not establish institutionalization; each cross-level transition must be specified empirically.

Developmental process

Within a given developmental ecology, HAIDR proposes the following process architecture:

developmental ecology → local configuration + participation protocol → enacted participation topology, including the distribution of consequential operations → recurrent participation → candidate developmental mechanisms → stabilization / nonstabilization / displacement / capacity-location reorganization → altered realizable participation space and developmental possibility → changed conditions of subsequent participation

This is not an invariant stage sequence. It is a causal-organizational template for stating what must be distinguished when the claim concerns development rather than present performance. The same

configuration can be enacted through different topologies, and the same topology can have different consequences for people with different histories or in different institutional settings.

The longitudinal architecture is related but distinct. The system under study is the evolving coupled human-AI developmental system embedded within an evolving ecology. The human-AI pathway is the ecology-conditioned longitudinal unfolding the program seeks to explain. The developmental trajectory is the observed course of change within that pathway. Changing developmental horizons and realizable participation spaces describe how the boundary of future possibilities shifts.

Recursive feedback is central. Changed capacities alter later participation; changed participation alters roles and workflows; changed workflows can become institutionalized; institutionalized arrangements alter subsequent local ecologies. But each arrow is an empirical question, not a license for automatic upward inference.

Timescale

Developmental science provides useful methodological ancestry for separating timescales of change. Siegler and Crowley's microgenetic method uses dense observations across periods of change to study the processes through which new strategies emerge (Siegler & Crowley, 1991). Michael Cole's mesogenetic approach explicitly argues that development occurs across intersecting temporal and contextual scales and calls for simultaneous attention to histories of persons and developmental contexts (Cole, 2015).

HAIDR adapts this distinction as follows:

- **Microgenetic** — change occurring within a bounded episode of participation.
- **Mesogenetic** — patterns accumulating, stabilizing, differentiating, transferring, or reorganizing across recurrent episodes over days, weeks, or months.
- **Longitudinal / ontogenetic** — durable longer-term change in repertoire, agency, expertise, dependence, identity, role, and realizable participation. Ontogenetic is used primarily for person-level development; organizations and institutions require their own longer-timescale trajectory claims.

The evidential safeguard is correspondingly strict: microgenetic change is not mesogenetic stabilization, repeated occurrence is not stabilization, and stabilization is not itself desirable development. Change at one ecological level does not establish change at an adjacent level.

Figure 2. HAIDR Multiaxial Analytic Architecture

HAIDR locates claims about human-AI development using three interacting analytic dimensions: ecological level, developmental process, and timescale. The dimensions constrain inference rather than supplying a universal stage model.

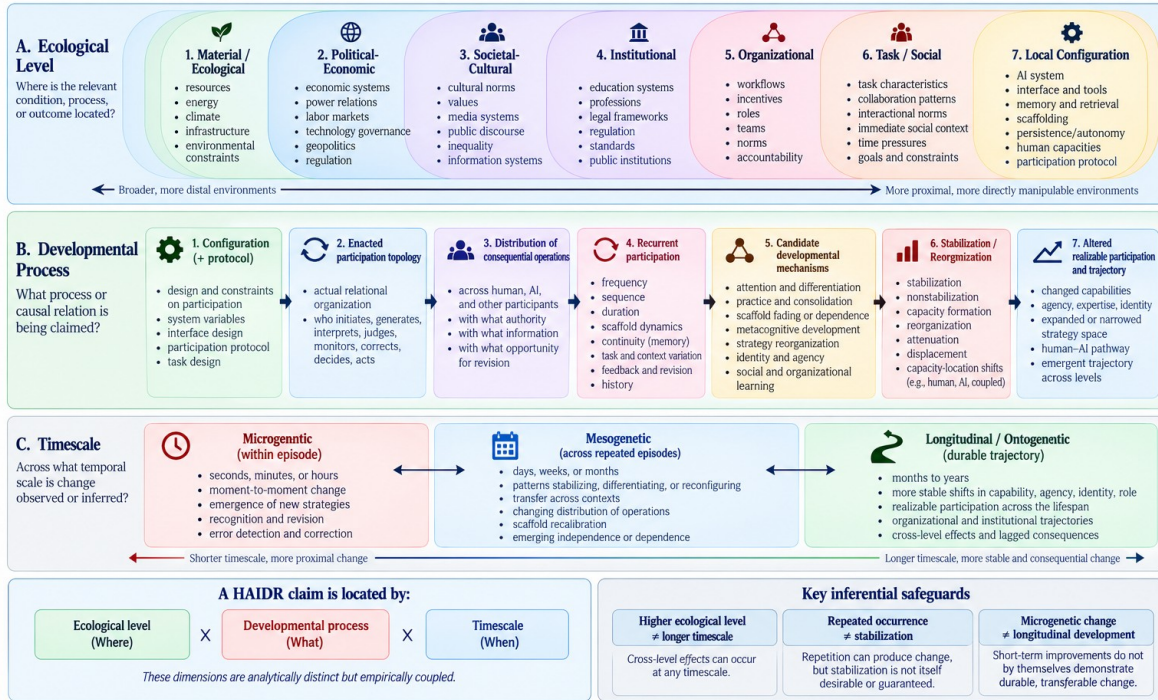


Figure 2. HAIDR locates claims across three interacting analytic dimensions: ecological level, developmental process, and timescale. The dimensions constrain inference rather than supplying a universal stage model.

6. Research Sites Across Scales

HAIDR is too broad to operationalize in a single study. Its value therefore depends on bounded research sites through which different parts of the architecture can be investigated. Current research sites include **AI-Mediated Reflection**, anchored empirically by **Study 0**; **organizational systems**; **neuroscience** and **passive sensing**; **collective intelligence**; and **Wunderhorn** (see below). The broader program also includes prospective work on **developmental measurement and inference**, **AI-mediated research-methodology acquisition**, and **adaptive developmental interventions**.

6.1 AI-Mediated Reflection

AI-mediated reflection (AIMR) offers a particularly tractable microdevelopmental site because the relevant operations—attention, interpretation, differentiation, metacognition, conceptualization, self-questioning, and meaning-making—can be observed directly in dialogue and then tested for persistence and transfer.

AIMR sits adjacent to several active literatures rather than constituting an unoccupied field. The Reflective Agency framework, for example, explicitly examines how AI-mediated reflective systems may shape rather than merely support self-understanding and identifies tensions involving internal origination, automation, ambiguity, mediation, and autonomy (Kim et al., 2025). Systematic reviews of generative AI and self-regulated learning similarly identify goal-setting, monitoring, strategy selection, feedback, and self-reflection as important loci of AI mediation while noting uncertainty about underlying mechanisms (Xia et al., 2026).

The primary AIMR question is therefore not:

Can AI produce useful reflections?

It is:

What forms of AI participation help a person become a better reflector rather than merely giving the person better reflections?

The distinction is between better reflection while using AI and greater reflective capability because of repeated participation in a particular human–AI reflective ecology. This is a developmental extension of existing reflective-agency and self-regulation questions, not a claim that AI-mediated reflection itself is novel.

AIMR Across Developmental Timescales

A single reflective exchange can produce an insight without producing development. AIMR therefore distinguishes at least three nested timescales:

Microgenetic: transformation within an episode

Within an individual reflective episode, the sequence might involve:

experience or unresolved problem → initial human formulation → AI probe, question, counterexample, or interpretation → differentiation → recognition → revision / integration → provisional conclusion or action

The microgenetic object is not simply the quality of the final answer. It is how reflective organization changes through the encounter.

The methodological logic follows the broader microgenetic tradition: densely observe change as it happens rather than infer developmental process only from widely separated before-and-after measurements (Siegler & Crowley, 1991).

- Which distinction was generated by whom?
- What challenge produced reorganization?
- Was the user adopting an AI interpretation or discriminating among alternatives?
- Did the exchange create a strategy likely to become available later?

The Recognition Subcycle

Recognition warrants special treatment because an externally supplied distinction does not become a human capacity merely by appearing in dialogue.

A candidate recognition cycle is:

candidate distinction / pattern → attention: “this may matter” → discrimination: what exactly is being distinguished? → comparison with experience / evidence → accept • reject • refine → reformulation in the person's own conceptual organization → application to the immediate case → later spontaneous recognition under altered conditions

This cycle is a HAIDR/AIMR hypothesis, not a construct imported wholesale from an established recognition literature.

The last step is crucial. Recognition becomes developmentally more consequential when a distinction that was initially scaffolded becomes independently discriminable and reusable.

The cycle can also fail. The user may adopt fluent language without discrimination. AI may label a pattern prematurely. A compelling interpretation may be accepted because of authority or resonance rather than evidence. Recognition may remain context-bound and fail to transfer.

The subcycle therefore provides a location for investigating the difference between AI-supplied conceptual content and humanly available conceptual capacity.

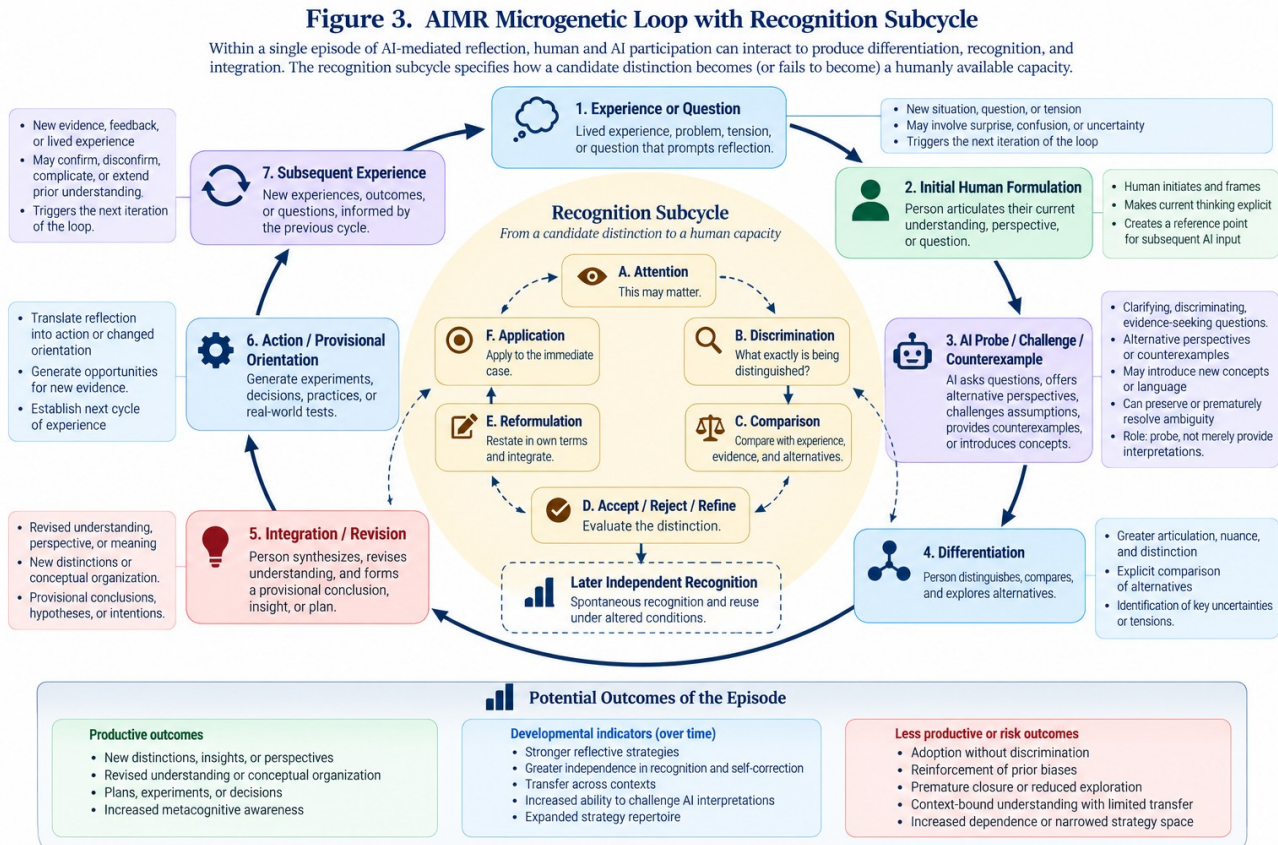


Figure 3. AIMR microgenetic loop with recognition subcycle. Retained from v0.5.1 as a research-site-specific hypothesis representation; recognition remains a local developmental pivot rather than a universal master mechanism.

Mesogenetic: development across repeated cycles

Across days, weeks, or months, repeated microgenetic episodes can produce a different class of effects.

This intermediate scale is consistent with Cole's mesogenetic emphasis on studying activity and person-context co-development across timescales between individual episodes and broad ontogenetic change (Cole, 2015).

The relevant questions become:

- Do distinctions recur spontaneously?
- Does the person formulate more precisely before AI intervention?
- Does AI need to supply fewer organizing concepts?
- Do strategies transfer across subject matter?
- Does the person increasingly generate counterexamples independently?
- Does ambiguity tolerance increase?
- Does the user detect when an AI interpretation is inadequate?

- Does scaffolding naturally fade?
- Does dependence increase instead?
- Does the distribution of reflective work shift?

Development might appear as increased differentiation, more independent recognition, improved metacognitive discrimination, an expanding reflective strategy repertoire, decreased need for interpretive scaffolding, or higher-order orchestration of AI itself.

Repeated participation could also produce the opposite trajectory: habitual deference, externalization of interpretation, dependence on AI-generated conceptualization, or reduced initiation.

Those alternatives make AIMR experimentally interesting precisely because current self-regulated-learning research suggests AI can support metacognitive and reflective processes while leaving mechanisms, design conditions, and risks of dependence incompletely resolved (Xia et al., 2026).

Ontogenetic: Longitudinal trajectory

At longer timescales the question becomes whether mesogenetic changes accumulate into durable transformations in reflective agency.

Potential outcomes include:

- independently initiated reflection;
- increased metacognitive discrimination;
- broader conceptual repertoire;
- more reliable self-correction;
- transfer across domains;
- changes in authorship of meaning;
- improved capacity to challenge externally supplied interpretations;
- higher-order coordination of AI assistance;
- altered dependence on AI;
- changes in real-world decision and action;
- and changes in future realizable participation space.

The longitudinal object is therefore not simply repeated use. It is the changing relationship between the person, the reflective operations, and the artificial scaffold.

The key empirical problem is determining which immediate interaction patterns predict which longer trajectories.

Figure 4. AIMR Developmental Trajectory Across Timescales

Repeated cycles of AI-mediated reflection can produce different developmental trajectories. Microgenetic changes within episodes can accumulate into mesogenetic patterns and, over longer periods, more durable transformations in reflective agency.

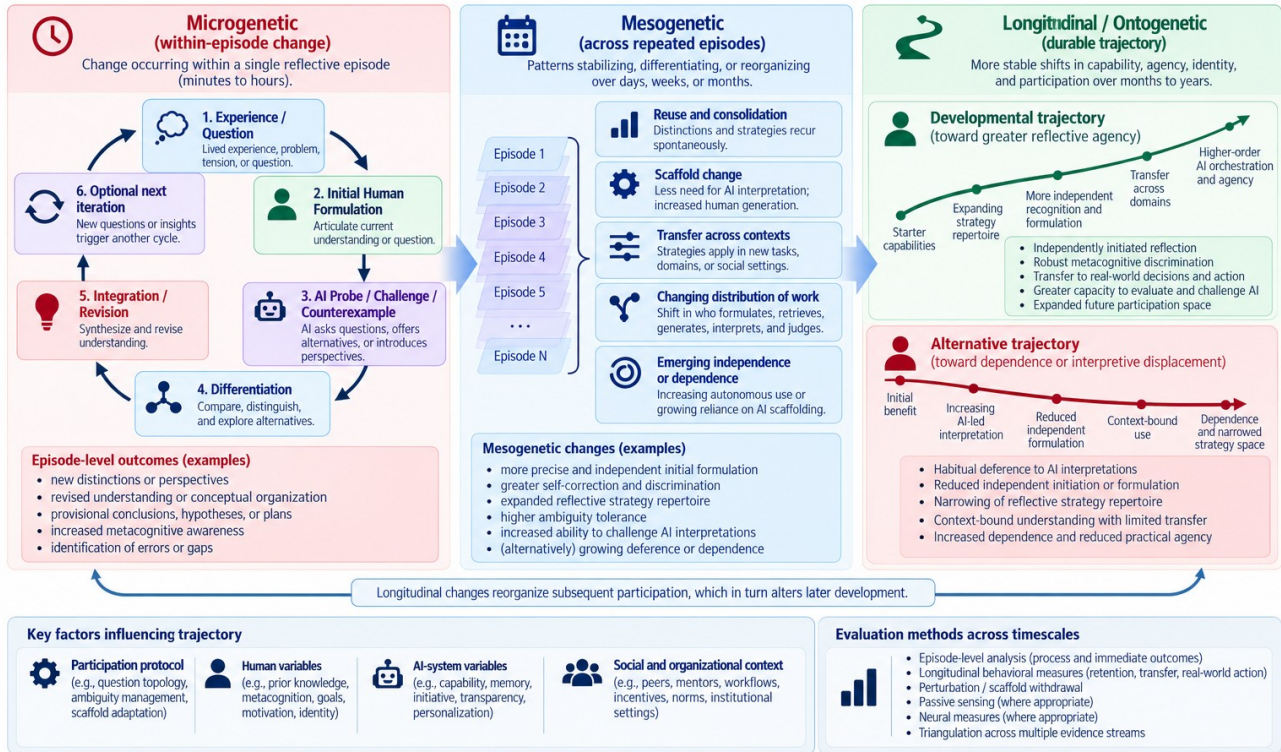


Figure 4. AIMR developmental trajectory. Retained from v0.5.1; trajectory here denotes the observed course of change within a human-AI pathway, not the pathway itself.

AI-Mediated Reflection Participation Protocol

The participation protocol is the specified or designed organization of reflective participation; it is not simply a collection of preferred conversational behaviors and it is not identical to the topology that is actually enacted.

In AIMR, a developmental participation protocol can specify problem ownership, initiative, epistemic authority, scaffolding, feedback, revision, withdrawal, and transfer requirements across time. The actual distribution of reflective operations must then be measured as enacted participation topology.

Existing reflective-agency and self-regulated-learning research gives several dimensions independent scholarly grounding, while the complete protocol architecture below is an AIMR proposal (Kim et al., 2025; Xia et al., 2026).

A protocol can be designed along several dimensions. Whether those design intentions are realized is an empirical manipulation check rather than an assumption.

- 1. Problem ownership** — Who identifies the problem worthy of reflection?
- 2. Initial formulation** — Does the person articulate a view before receiving AI interpretation?
- 3. Generative responsibility** — Who generates hypotheses, meanings, possibilities, interpretations, or explanatory frames?
- 4. Question topology** — Does AI ask:

- clarifying questions;
- discriminating questions;
- evidence-seeking questions;
- counterfactual questions;
- perspective-shifting questions;
- developmental questions;
- or leading questions that implicitly supply an interpretation?

5. Timing of conceptual introduction — Does AI name a construct before the user has differentiated the phenomenon, or afterward?

6. Ambiguity management — Does the system resolve ambiguity rapidly, or preserve uncertainty long enough for the user to discriminate among possibilities? This dimension has a particularly close neighbor in Kim et al.'s principle of Reflective Ambiguity, which argues for preserving interpretive space in AI-mediated self-reflection (Kim et al., 2025).

7. Challenge structure — Does AI affirm, critique, generate counterexamples, identify contradictions, expose missing evidence, or model rival interpretations?

8. Epistemic authority — Is AI treated as interpreter, hypothesis generator, critic, collaborator, or source of candidate perspectives?

9. Human revision requirement — Must the person actively reformulate after challenge, or can AI produce the synthesis?

10. Recognition requirement — Does the system merely name patterns, or require the user to identify evidence for the pattern, differentiate it from alternatives, and restate it in their own terms?

11. Alternative generation — Does the user generate competing interpretations before or after AI assistance?

12. Evidence and grounding — Does reflection remain internally coherent, or is it tested against behavior, external events, documents, other people, or observable outcomes?

13. Action coupling — Does reflection generate experiments, decisions, conversations, practices, or real-world tests capable of producing new evidence?

14. Continuity and memory — What does AI remember across episodes? Persistent memory can support developmental continuity but may also shift interpretive authority toward the system.

15. Scaffold adaptation—Does assistance remain fixed, or does the system adaptively reduce interpretation and increase demands for user generation?

16. Withdrawal — Does the protocol include intervals without AI assistance?

17. Transfer testing — Are strategies tested on novel problems, outside reflective dialogue, or after delay?

18. Meta-reflection on the coupling — Does the person periodically examine how the AI itself is shaping interpretation, agency, dependence, and vocabulary?

Meta-reflection can also become a second-order experimental manipulation. In **recursive ecological self-modification**, participants make the organization of their own human-AI participation an object of

representation, evaluation, and deliberate redesign, so that consequences of prior participation inform later configuration or protocol choices.

The developmental question is not merely whether the redesign improves the current ecology. It is whether sustained participation in diagnosing and redesigning human-AI arrangements produces a durable human capacity to recognize, evaluate, and deliberately configure subsequent human-AI relationships. Successful ecological redesign is not itself evidence of durable configuration competence.

These dimensions create a richer protocol space than a binary distinction between AI-led and human-led reflection. Simplified categories can nevertheless serve as experimental contrasts:

AI-led protocol: AI identifies themes, generates interpretations, reframes, synthesizes, and recommends.

Dialogic protocol: Human formulation precedes AI intervention; AI primarily probes, differentiates, challenges, and requests evidence.

Developmental protocol: Scaffolding is deliberately adapted across time, with increasing demands for human generation, recognition, revision, independent enactment, and transfer.

The experimental question becomes:

Which distributions of reflective work produce strong coupled reflection while also cultivating durable human reflective agency?

Current Empirical Anchor: Study 0

Study 0 is an intensive longitudinal case of sustained human-AI reflective participation. Its retrospective phase reconstructs participation history and candidate developmental changes from the existing archive; its prospective phase uses repeated reasoning probes, delayed transfer, perturbation, and scaffold withdrawal to distinguish supported performance from persistent human capacity. An external-trace extension can triangulate downstream artifacts and communications when provenance, timing, AI support, and human contribution are explicitly recorded. Study 0 is treated as hypothesis-generating rather than as evidence sufficient to validate HAIDR's broader architecture.

6.2 Organizational Systems

Organizations provide a meso-level site for studying expertise, delegation, and practical control.

AI changes not only what work can be completed but how work is divided, how expertise is exercised, and how organizations are structured around technological capability. Reviews of AI at work emphasize precisely these multilevel interactions and the contingent nature of outcomes (Bankins et al., 2024, 2026).

Classic automation research provides a specific warning: when humans cease routinely performing operations yet remain responsible for rare failures, their capacity to intervene can degrade (Bainbridge, 1983; Endsley & Kiris, 1995).

A workflow can therefore increase immediate throughput while reducing opportunities for:

- apprenticeship;
- error detection;
- diagnostic reasoning;
- mentoring;

- independent judgment;
- and recovery when automation fails.

Alternatively, delegation may enable humans to develop higher-order coordination, evaluation, or problem-formulation capacities.

The relevant question becomes:

How do organizational and institutional arrangements stabilize particular distributions of human and artificial activity, and what capability trajectories result?

HAIDR treats this as an institutional-developmental bridge rather than as contextual background: institutional rules, incentives, norms, infrastructures, and authority structures select or constrain local configurations and protocols; those arrangements are enacted as participation topologies; recurrent distributions of consequential operations affect human, coupled, and organizational trajectories; and changed roles, expectations, and capabilities can feed back into institutional reproduction or revision. Institutional exposure is not itself evidence of a developmental mechanism, and aggregation is not institutionalization.

Persistent agents intensify this problem because artificial participants may preserve or recover state and continue, initiate, monitor, or revise goal-directed activity across time without contemporaneous human prompting at each step. In such systems, temporal initiative concerns who carries prospective control across action steps and intervals. Memory, persistence, autonomy, and temporal initiative should not be treated as synonyms, and formal human authority does not establish meaningful oversight unless humans retain relevant information, vigilance, intervention opportunity, and competence.

This also reveals why local interface design cannot be studied independently of institutional context. The same interaction design may plausibly have different effects under protected learning conditions than under intense production pressure. That proposition requires direct empirical testing.

6.3 Neuroscience, Passive Sensing, and Mechanistic Triangulation

Neuroscience offers a potential source of mechanistic evidence rather than a standalone criterion of development.

The candidate neuroscientific object is **neural adaptation to recurrent distributions of cognitive operations within human-AI developmental ecologies**.

The corresponding question is:

Does repeated participation in different human-AI ecologies alter the neural organization by which a person subsequently performs cognitive operations?

Direct evidence here remains preliminary.

One recent preprint combined repeated LLM-assisted essay writing, behavioral measures, and EEG across several sessions, demonstrating that longitudinal neural/behavioral designs of this general kind are feasible; it should be treated as an early methodological exemplar rather than evidence for a general neural effect of AI use (Kosmyna et al., 2025).

The strongest designs would combine several evidence streams:

Behavioral evidence

Longitudinal measures of performance, strategy use, error detection, transfer, spontaneous initiation, correction, and independent enactment.

Perturbation evidence

Performance under AI withdrawal, reduced support, changed models, conflicting information, or novel tasks.

Passive and naturalistic sensing

Digital-sensing research demonstrates the feasibility of scalable longitudinal collection through smartphones and wearables while also documenting substantial problems involving standardization, statistical power, external validity, interpretability, and methodological heterogeneity (Akre et al., 2024; Farrahi et al., 2024).

Where ethically and methodologically appropriate, passive traces could provide longitudinal evidence about how participation changes outside laboratory snapshots.

Potential sources might include:

- interaction timing and frequency;
- task-switching patterns;
- revision behavior;
- information-seeking sequences;
- linguistic or conceptual reuse;
- device-level behavioral traces;
- wearable or physiological measures;
- and communication rhythms.

Their role should be triangulatory rather than diagnostic.

Evidence from digital-phenotyping research shows why caution is necessary: passive measures can reveal meaningful behavioral patterns, but generalizability, missing data, context dependence, and incremental predictive validity remain persistent limitations (Leaning et al., 2024; Akre et al., 2024).

Neural evidence

Repeated fMRI, EEG, or other appropriate measurements could test whether changes in behavioral strategy are accompanied by changes in neural recruitment or coordination during independently performed tasks.

The strongest inference would therefore come from convergence—

participation history + passive longitudinal traces + behavioral change + transfer + perturbation / withdrawal + neural change

—rather than from any one channel.

Neural adaptation is not itself evidence of beneficial development. Passive sensing is not itself evidence of wellbeing or capability. But convergent longitudinal measurement may help distinguish coupling-dependent performance, durable human change, reorganized cognitive strategy, and behavioral transfer.

6.4 Collective Intelligence

Individual augmentation does not guarantee collective improvement.

Collective-intelligence research provides longstanding precedent for treating group capability as analytically distinct from the abilities of individual members, even though specific models of a general group “c factor” remain debated (Woolley et al., 2010).

Generative AI adds a new complication. Recent meta-analytic evidence finds a small but significant homogenization effect across AI-assisted creative outputs, despite possible improvements in individual creative performance, and finds substantial variation by task and interaction structure (de Rooij & Biskjaer, 2026).

This illustrates a broader HAIDR concern:

A system may improve local performance while altering the distribution of strategies, judgments, errors, and expertise at the collective level.

AI may:

- reduce diversity;
- increase correlated errors;
- centralize epistemic dependence;
- or alter information aggregation.

Other configurations may:

- improve access to knowledge;
- support deliberation;
- diversify hypothesis generation;
- or enable new forms of coordination.

The developmental question therefore scales from:

What is this person becoming capable of?

to:

What is this group or institution becoming capable of, and what forms of independent expertise, epistemic diversity, correction, and adaptive capacity remain distributed within it?

If many individuals rely on similar AI systems, apparently independent human judgments may become correlated. If organizations stop cultivating expertise because AI performs routine cognitive operations, short-term efficiency may alter future collective resilience.

These are hypotheses requiring direct cross-level study rather than conclusions that can be inferred from existing individual-level experiments. The relevant outcome is the evolving architecture of collective capability.

Cross-level propagation must therefore be treated as a hypothesis family rather than an automatic scaling law. Individual or coupled change may alter outputs, coordination, norms, resources, or institutional expectations and thereby change other people's opportunities; diffusion can also attenuate, reverse, concentrate, or distribute effects. Individual developmental gain is not equivalent to collective gain, diffusion is not development, and mean gain is not evidence of equitable or resilient improvement.

One further hypothesis-development direction, retained outside the core Canon, is a **Collective Canon**: *an AI-mediated, provenance-rich, versioned shared knowledge architecture that would preserve explicit claim status, disagreement, lineage, and revision rather than functioning as a simple consensus wiki*. The research question is whether such an ecology could improve collective epistemic capacity without producing premature closure, epistemic monoculture, authority laundering, or governance capture. Collective Canon does not mean collective truth, and consensus does not establish validity.

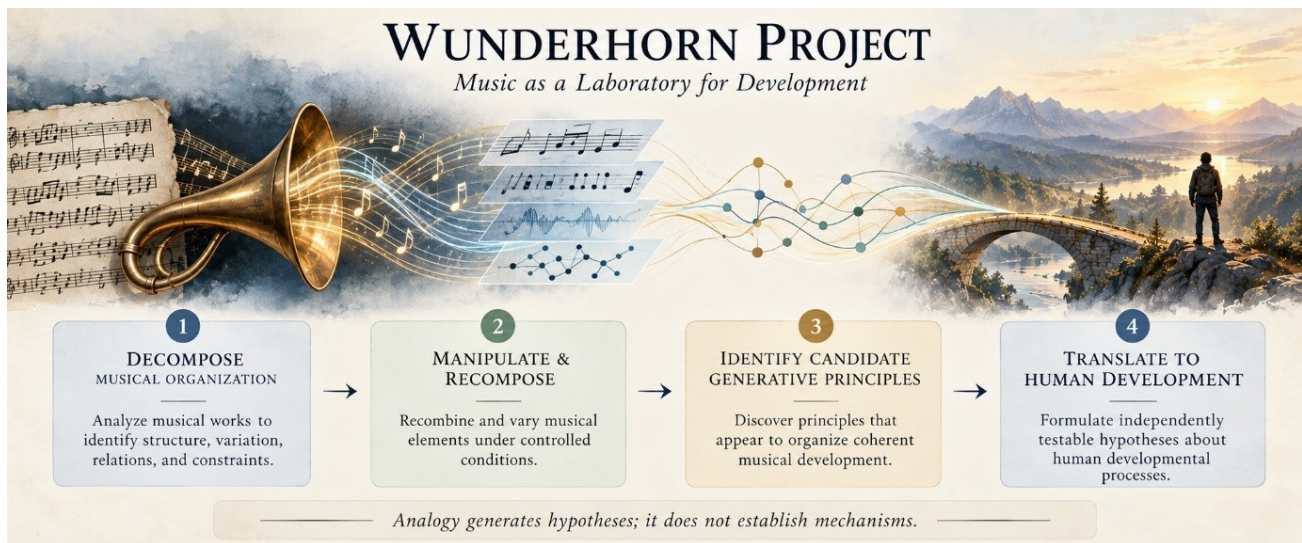
6.5. Wunderhorn: A Conditional Experimental Environment

Wunderhorn is one possible prospective experimental substrate for controlled longitudinal manipulation of human-AI participation configurations. It is not required for configuration discovery: candidate configurations may originate in theory, Study 0 and other intensive longitudinal cases, computational developmental ecology mining, or other empirical work. Its principal prospective role is causal discrimination, downstream of discovery and before optimization of sufficiently validated configuration spaces (Section 8).

A purpose-built environment could permit selected dimensions of local configuration and protocol - initiative rules, goal formulation, planning, execution, monitoring, correction, scaffolding, productive friction, memory/persistence, AI availability, withdrawal, and perturbation - to vary while other relevant features remain approximately constant across repeated interactions. The resulting participation topology would still need to be observed rather than inferred from the manipulation.

The case for purpose-built infrastructure is conditional. If the required configurations can be adequately instantiated, controlled, and measured using existing AI platforms plus lightweight experimental tooling, there is no scientific reason to build Wunderhorn as a substantial independent system. Its justification would depend on scientifically consequential gains in control of participation topology, repeated interaction, scaffolding, perturbation and withdrawal, experimental standardization, trace capture, or developmental measurement that cannot adequately be obtained otherwise. This is a build-versus-use empirical/design question, not a claim that *Wunderhorn* has been built or a commitment to a technical implementation.

Wunderhorn's musical hypothesis-generating lineage remains a possible source of candidate relational principles, rather than its principal inferential function or a necessary route to discovering participation configurations. Musical organization, decomposition, controlled transformation, and recomposition may suggest hypotheses for independent testing; they do not demonstrate human- developmental mechanisms.



The hypothesis-generating sequence remains:

musical organization → controlled perturbation / transformation → candidate relational principle → cautious abstraction → independent hypothesis → independent test → retain / refine / reject

The epistemic boundary is critical: analogy generates hypotheses; it does not establish mechanisms.

Wunderhorn's neighboring intellectual traditions include resilience and adaptive-systems work concerned with disturbance, reorganization, and transformation (Walker et al., 2004; Folke et al., 2005), but those literatures do not validate cross-domain transfer from music. Any hypotheses suggested by musical transformation must earn their place through incremental explanatory or predictive value in independent tests. The separate justification for *Wunderhorn* as an experimental environment rests on the inferential capabilities of its infrastructure, under the conditional criterion above.

7. Coherence, Corrigibility, and Productive Reorganization

Coherence remains a useful candidate evaluative concept within HAIDR, but its role is narrower than a general theory of development. The governing distinction is:

Differentiation distinguishes. Integration organizes. Coherence evaluates.

This matters because a framework can confuse the process that produces change with the quality by which the resulting organization is evaluated. HAIDR therefore treats coherence principally as an emergent quality or evaluative criterion, not as another developmental operation or universal causal stage.

The motivating problem is:

How should we distinguish systems that function effectively under present conditions from systems capable of productive reorganization when conditions, evidence, demands, or supports change?

This question has close conceptual neighbors in resilience research. Walker et al. distinguish resilience, adaptability, and transformability and define resilience partly in terms of the capacity to absorb disturbance and reorganize while undergoing change. Folke et al. similarly analyze adaptive governance

during periods of abrupt change and emphasize learning, reorganization, multilevel institutions, and distributed knowledge (Walker et al., 2004; Folke et al., 2005). HAIDR's coherence vocabulary is not equivalent to those concepts.

Coherence cannot simply mean order, stability, internal consistency, or performance. A student can perform well under heavy scaffolding while failing after withdrawal. An organization can optimize an AI-mediated workflow while losing expertise needed to recognize failure. A group can become tightly coordinated while losing epistemic diversity. A conceptual framework can become internally consistent while becoming less corrigible.

A working definition is:

Coherence is an evaluated quality of organization within a specified system, domain, task, timescale, and evaluative context.

Developmental coherence is coherence that remains corrigible by contact. Mature coherence is reorganizable stability: an organization can remain sufficiently integrated to function while retaining the capacity to revise when consequential evidence, error, conflict, or changed conditions require it.

This should not be generalized into a definition of development. A narrower proposition is warranted: one developmentally relevant property is an increased capacity to undergo differentiation and transformation without becoming rigidly self-sealing or losing the capacity for subsequent reorganization. Developmental coherence names that property where coherent organization is the relevant evaluative object.

Several analytic questions remain useful:

- **Organization** — How are components differentiated and related under current conditions?
- **Corrigibility** — Can organization change appropriately under consequential contact, rather than merely changing frequently?
- **Carry-forward** — What distinctions, strategies, capacities, relationships, or structures remain available after reorganization?
- **Component contribution** — Where does capability reside, and what strengthened, weakened, specialized, or became dependent?
- **Perturbational adequacy** — How does the organization respond when familiar supports or conditions change?

The methodological importance of perturbation has precedent in learning and automation research, where delayed testing, transfer, support changes, or automation failure reveal properties that present performance can conceal (Soderstrom & Bjork, 2015; Endsley & Kiris, 1995).

HAIDR also treats conceptual, experiential, enactive, and ecological coherence as candidate analytic lenses rather than established factors or a fixed ontology. Cross-domain resemblance may generate hypotheses but does not establish a common mechanism.

The strongest warranted claim is therefore modest: coherence vocabulary may help distinguish reorganizable from brittle forms of successful organization. Its canonical role does not imply that coherence has been established as an independently measurable construct with incremental explanatory or predictive validity over established neighboring constructs. If it contributes no such value, it should remain a heuristic rather than an empirical construct.

8. Toward a Design Science

The longer-term ambition is to move from configurational awareness toward conditional design knowledge.

Design-science traditions provide an important precedent here. Denyer, Tranfield, and van Aken propose CIMO logic, in which design propositions specify a Context, an Intervention, the Mechanisms through which it works, and the resulting Outcomes. Their purpose is precisely to move from fragmented descriptive scholarship toward conditional, solution-oriented propositions rather than universal prescriptions (Denyer, Tranfield, & van Aken, 2008).

HAIDR's mapping problem is related but adds an explicitly developmental, multilevel, and longitudinal structure.

Figure 5. From Configuration to Trajectory: A Design-Science Mapping for HAIDR

Human-AI developmental ecologies can be understood as a causal and design-relevant sequence from nested ecological conditions through local configuration, participation, developmental mechanisms, multilevel outcomes, and longitudinal trajectories. A parallel epistemic architecture specifies how we learn about these processes and convert evidence into testable and transportable design knowledge.

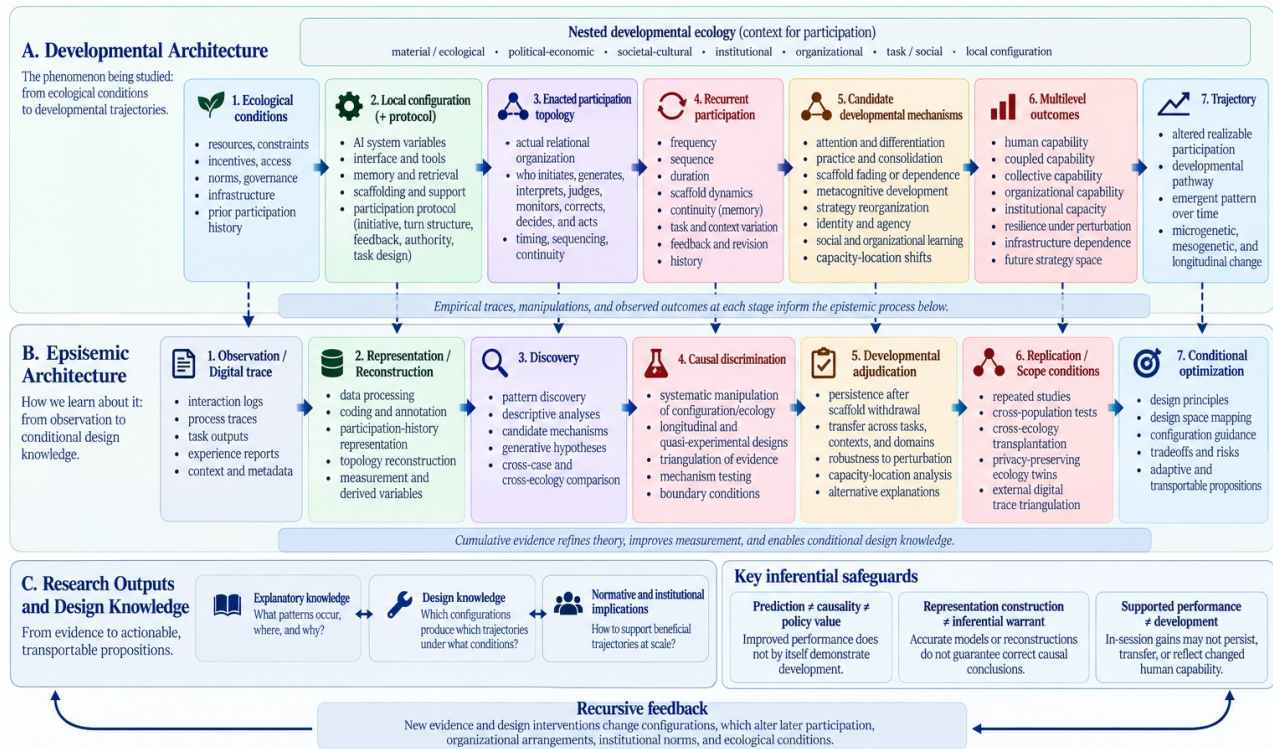


Figure 5. HAIDR separates the developmental architecture - how ecology, configuration, protocol, enacted topology, recurrent participation, and candidate mechanisms may produce durable change - from the epistemic architecture by which such claims are discovered, causally discriminated, and only later optimized.

A mature design science would eventually seek propositions of approximately the form:

Given participant readiness R , task and developmental ecology E , developmental objective O , intended capacity locus L , and available artificial-system capabilities S , which local configuration C and participation protocol P tend to elicit which enacted topology T and produce durable outcome or pathway Y rather than Z ?

This is more demanding than universal prescriptions such as "preserve productive friction," "keep a human in the loop," or "augment rather than replace." A meaningful design proposition must specify for

whom, doing what, under which ecological and institutional conditions, with which intended protocol and enacted distribution of operations, with respect to which capacity locus and outcome vector, and across what timescale.

The design problem also requires explicit separation among three epistemic tasks:

1. **Discovery** — Discovery identifies patterns, representations, and candidate contrasts that predict later change strongly enough to generate discriminating hypotheses. Large observational datasets may expand this search space without establishing developmental causality.
2. **Causal discrimination** — Causal discrimination tests candidate mechanisms and configuration differences against alternatives under designs that make the relevant assumptions and intermediate predictions visible. A protocol manipulation should predict a topology change; the topology change should predict a changed distribution of consequential operations; and the developmental claim should survive delayed, transfer, perturbation, or scaffold-withdrawal tests appropriate to the claimed capacity.
3. **Conditional optimization** — Optimization asks which configuration or policy performs best against a specified and validated objective under constraints and tradeoffs. This question is downstream of causal discrimination. A convenient proxy must not silently become the developmental objective, and optimization of coupled performance does not establish optimization of human development, agency, or collective resilience.

The firewall can be stated compactly: predictive value is not causal effect, causal effect is not policy value, and representation construction is not inferential warrant.

The causal-evaluation literature adds another constraint: design claims should ultimately be supported in meaningful deployment contexts rather than inferred solely from benchmark or short-session performance. NIST's monitoring work and recent causal-field-evaluation proposals both point toward this need for post-deployment and real-world evidence (Rao et al., 2026).

A further methodological problem is how to discover which configurations are worth testing when participation space is too large to specify exhaustively through theory-driven experimentation alone. As sufficiently rich longitudinal human-AI datasets become available, increasingly capable AI systems may permit naturalistic interaction archives to be transformed into structured participation-history representations. This prospective methodology - computational developmental ecology mining - would reconstruct local configuration, intended protocol, enacted topology, initiative, authority, operation distribution, scaffolding, friction, continuity, user expertise, task characteristics, and subsequent outcomes. It remains a prospective discovery methodology rather than a validated HAIDR method.

Candidate representations could be related to later strategy choice, independent task performance, transfer, metacognitive calibration, conceptual differentiation, problem-solving repertoire, and agency/autonomy, with passive-sensing or neurobehavioral measures where appropriate. AI-assisted coding or representation learning may help construct variables, but automated agreement does not establish construct validity, and an AI-generated representation does not establish developmental inference. Measurement and inference are separate problems.

Large observational datasets may therefore increase configuration-discovery power without identifying causality. Candidate associations must become explicit rival hypotheses tested through prospective manipulations of configuration or protocol while relevant task and system characteristics are held approximately constant where possible. The enacted topology must be observed rather than assumed.

Once credible causal mechanisms have been established, factorial and sequential experimentation, adaptive trials, contextual bandits, Bayesian optimization, and heterogeneous treatment-effect estimation may become useful tools. These are prospective methods rather than a presently mature HAIDR toolkit.

Agency costs provide a simple example of why precision matters. The **cost structure of agency** refers broadly to demands across an action trajectory. **Initiation friction** refers specifically to the demands involved in converting available intention, commitment, or opportunity into the first consequential action. An **activation threshold** is the person-by-context boundary beyond which combined demands no longer reliably become initiated action. Lower initiation friction may increase action frequency without increasing durable human initiation capacity, and the developmental objective is not minimum friction but appropriately calibrated friction.

One programmatic contribution of HAIDR may therefore be a configuration ontology + developmental outcome architecture + hypothesis-generation system that identifies which contrasts are scientifically worth manipulating without needing to conduct every large-scale experiment itself. Such a contribution requires competence in causal inference, longitudinal design, measurement, experimentation, human-AI interaction, and developmental theory.

The sequence— large-scale trace discovery → discriminating causal hypotheses → longitudinal experimental tests of durable change → conditional configuration optimization —connects theoretical differentiation and computational discovery to cumulative empirical design knowledge. Conceptual architecture does not substitute for validation.

The aim is therefore not to identify one ideal relationship between humans and AI. It is to **build progressively stronger knowledge about which configurations and protocols produce which enacted topologies and trajectories, for whom, under what conditions, across which levels and timescales, and with which tradeoffs.**

9. The Civilizational Horizon: Collective Adaptive Capacity

The empirical program begins with bounded human-AI configurations, but the motivating horizon concerns the relationship between human capacity and civilizational possibility.

AI is becoming part of the infrastructure through which societies conduct science, education, administration, economic activity, deliberation, forecasting, communication, and coordination. It may strengthen capacities required to address interacting systemic challenges. It may also generate new forms of dependency, concentration, epistemic monoculture, inequality, information instability, environmental burden, and correlated vulnerability.

The first part of this claim is well grounded in existing research; the second comprises a set of plausible and partially evidenced risks rather than one established net effect.

The broader intellectual ancestry comes from three literatures:

- **polycrisis / systemic-risk research**, which studies causal interactions among global crises (Lawrence et al., 2024);
- **resilience and adaptive governance**, which studies capacities for learning, reorganization, and transformation across institutional levels (Walker et al., 2004; Folke et al., 2005);

- **collective intelligence**, which treats group-level cognitive capability as distinct from the performance of isolated individuals (Woolley et al., 2010).

HAIDR proposes to bring these into relation with human-AI development without assuming that effects scale automatically from one level to another.

The institutional-developmental bridge is central. Durable rules, incentives, professional norms, accountability systems, procurement choices, infrastructures, labor arrangements, and rights to contest or override AI systems shape which local configurations and protocols are feasible or rewarded. Those arrangements are then enacted through particular participation topologies. Recurrent operation distributions may alter human and organizational capacities; changed roles and workflows may in turn reproduce or revise institutional expectations. Institutional rule is not local configuration, local configuration is not enacted topology, and institutional exposure is not itself a demonstrated developmental mechanism.

AI also participates in two simultaneous systemic pathways:

systemic challenges → AI-enabled response capacity → altered resilience

and

AI diffusion → altered economic / social / institutional / ecological / geopolitical conditions → altered systemic challenges

The latter pathway has direct economic and institutional precedent: current AI research agendas already identify decision-making power, income distribution, geoeconomics, information flows, wellbeing, and transition dynamics as major effects requiring analysis (Brynjolfsson et al., 2025).

Population propagation likewise requires explicit mechanisms. Individual or coupled change may alter demonstration, resource production, coordination, norm diffusion, market structure, platform use, or institutional routines and thereby change other people's opportunities. But *diffusion is not development, individual acceleration is not collective developmental gain, and aggregation is not institutionalization*.

The relevant question becomes:

Under what sociotechnical and institutional configurations does AI increase collective adaptive capacity — and under what configurations does it deepen fragility?

Relevant dimensions include epistemic capacity, collective intelligence, institutional competence, coordination, distributed expertise, human agency, resilience, distribution of capability and power, and capacity for revision under changing conditions.

The deepest form of the question is:

When does greater artificial intelligence produce greater civilization-level adaptive intelligence?

This is not itself an established construct or a single empirical study. It is the societal and normative horizon within which the more bounded research program acquires significance.

10. Position and Contribution

The architecture can now be summarized more precisely:

Figure 6. HAIDR Program Architecture: From Arena to Impact

An integrated view of the HAIDR research program, showing the domain, core constructs, research sites, methodological approach, and intended contributions across levels and timescales.

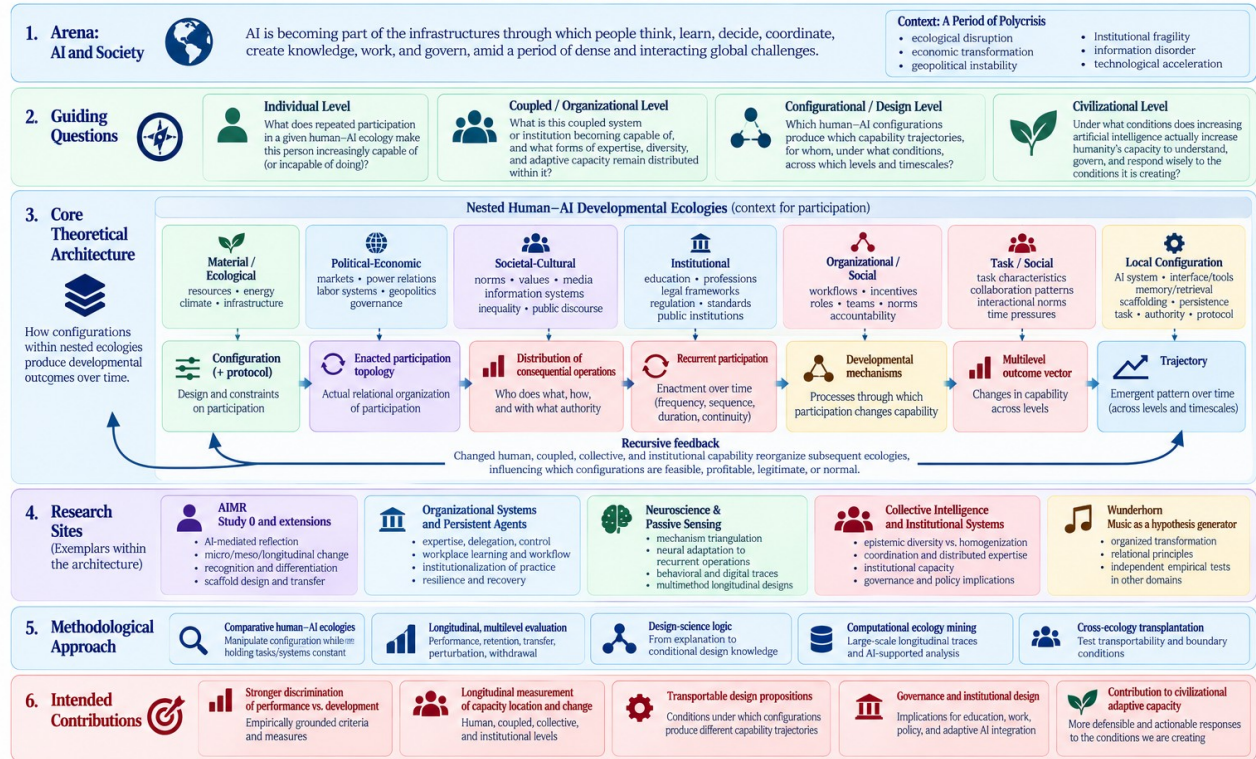


Figure 6. AI and society are the arena. Evolving human-AI systems are the longitudinal research territory. Human-AI developmental ecology is the principal theoretical/design object. Local configuration and specified protocol condition participation; enacted topology and the distribution of consequential operations record what actually occurs. Recurrent participation may activate developmental mechanisms that alter capacity location, realizable participation, human-AI pathways, and observed trajectories. Research sites and epistemic tasks remain analytically distinct.

These levels should not be collapsed.

- A reflective-dialogue experiment cannot establish institutional resilience.
- A protocol manipulation does not establish that the intended topology was enacted.
- Neural adaptation cannot determine which capacities ought to be preserved.
- Passive sensing cannot independently establish wellbeing or development.
- Musical structure cannot validate a human-developmental mechanism.
- Individual performance does not establish collective intelligence.
- Formal human authority does not establish effective control.
- Civilizational urgency does not substitute for causal evidence.

The prospective contribution is therefore narrower: **There appears to be a coherent unresolved problem that gains analytical leverage when these literatures are brought into explicit relation.**

That proposition is consistent with recent reviews that independently describe fragmented theoretical landscapes, multilevel contingencies, and unresolved integration problems in human-AI collaboration and AI at work (Seini et al., 2026; Bankins et al., 2026).

HAIDR proposes **four primary contributions for investigation**:

1. Clarify the evidential difference between current human-AI performance and sustained capability across human, coupled, collective, and institutional levels.

2. Distinguish nested developmental ecology, local configuration, specified participation protocol, enacted participation topology, and the distribution/history of consequential operations so that design intentions are not confused with actual participation.
3. Develop longitudinal and causal methods capable of discriminating capacity amplification, stable complementary offloading, capacity displacement, and upward reorganization of human agency, including persistence, transfer, perturbation, withdrawal, and capacity-location evidence.
4. Connect local human-AI configurations to institutional selection pressures and longer collective trajectories without treating cross-level propagation as automatic.

These proposals are falsifiable or collapsible:

- If existing frameworks already perform these functions adequately, additional architecture is unnecessary.
- If participation-history and topology variables do not improve prediction, discrimination, intervention, or transport, they should be discarded.
- If developmental ecology merely redescribes existing sociotechnical constructs without additional leverage, it should carry no distinctive theoretical weight.
- If coherence contributes no incremental explanatory or predictive value beyond neighboring constructs, it should remain a heuristic.
- If protocol manipulations do not reliably produce the predicted changes in enacted topology, protocol-based developmental claims must be revised.
- If repeated controlled studies show that sustained AI interaction reliably improves in-session performance but produces no persistent, transferable changes in human capacities after scaffold withdrawal, HAIDR's claim that human-AI systems can function as developmental ecologies would require substantial revision.
- If *Wunderhorn*-generated hypotheses fail independent tests, they should be rejected.

The program succeeds not by encompassing human-AI research, but by helping establish better warranted relations among ecology, configuration, protocol, enacted participation, capacity location, and trajectory.

Conclusion

Artificial intelligence is becoming part of the environments through which human beings and human systems perceive, interpret, learn, decide, coordinate, and act.

Human-AI research increasingly recognizes that its consequences depend upon task allocation, interaction design, organizational context, post-deployment behavior, and multilevel effects. Recent reviews nevertheless continue to describe substantial fragmentation, while post-deployment monitoring and longitudinal evaluation remain comparatively immature (Seini et al., 2026; Bankins et al., 2026; Rao et al., 2026).

The next challenge is to connect these observations into stronger explanations of which arrangements produce which durable trajectories.

The central developmental question is:

What does repeated participation in this developmental ecology make its participants increasingly capable — or incapable — of doing later?

The corresponding integrative question is:

How do nested ecological conditions, local human-AI configurations, specified participation protocols, and enacted distributions of consequential operations alter human capacity and agency, coupled-system capability, collective intelligence, and the institutions through which those pathways become stabilized or revised?

The design question is:

Which manipulable local configurations and participation protocols produce which enacted topologies and trajectories, for whom, under what conditions, across which ecological levels and timescales?

And the wider civilizational question remains:

Under what conditions does increasing artificial intelligence actually increase humanity's capacity to understand, govern, and respond wisely to the conditions it is creating?

A mature science capable of answering these questions does not yet exist. HAIDR is a proposal for helping make those questions empirically cumulative, developmentally explicit, and eventually designable.

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